1. User Management Module

Entity Name	Attributes	Key Type	Description
User	`user_id` (UUID)	**PK**	Unique user identifier
	`username` (VARCHAR)		Login name
	`email` (VARCHAR)		Contact email
	`password_hash` (VARCHAR)		Encrypted password
	`role_id` (UUID)	**FK**	Links to **Role** table
	`status` (ENUM)		Active, Inactive, Suspended
	`created_at` (TIMESTAMP)		Account creation date
Role	`role_id` (UUID)	**PK**	Unique role identifier
	`name` (VARCHAR)		e.g., "Admin", "Auditor"
	`permissions` (JSON)		Specific access rights

2. Document Management Core

Entity Name	Attributes	Key Type	Description
Folder	`folder_id` (UUID)	**PK**	Unique folder ID
	`name` (VARCHAR)		Folder name
	`parent_folder_id` (UUID)	**FK**	Self-reference for nesting
	`path` (VARCHAR)		Materialized path (e.g., "/docs/finance")
Document	`document_id` (UUID)	**PK**	Unique document ID
	`title` (VARCHAR)		Document name
	`folder_id` (UUID)	**FK**	Links to **Folder**
	`owner_id` (UUID)	**FK**	Links to **User**

Entity Name	Attributes	Key Type	Description
	`current_version_id` (UUID)		Reference to latest version
DocumentVersion	`version_id` (UUID)	**PK**	Unique version ID
	`document_id` (UUID)	**FK**	Links to **Document**
	`version_number` (VARCHAR)		e.g., "1.0", "1.1"
	`s3_bucket_key` (VARCHAR)		File path in AWS S3
	`checksum` (VARCHAR)		For integrity check
DocumentMetadata	`metadata_id` (UUID)	**PK**	Unique metadata entry
	`document_id` (UUID)	**FK**	Links to **Document**
	`key` (VARCHAR)		e.g., "Invoice Number"
	`value` (VARCHAR)		e.g., "INV-2025-001"

Relationships:

- **User (1) ↔ (N) Document**: A user owns many documents.
- **Folder (1) ↔ (N) Document**: A folder contains many documents.
- **Document (1) ↔ (N) DocumentVersion**: One document has history of many versions.
- **Document (1) ↔ (N) DocumentMetadata**: One document has many custom fields.

3. ERP Integration Engine

Entity Name	Attributes	Key Type	Description
ERPConnection	`connection_id` (UUID)	**PK**	Config for an ERP instance
	`name` (VARCHAR)		e.g., "SAP Production"
	`api_endpoint` (VARCHAR)		URL for the ERP API
	`auth_config` (JSON)		API Keys/Tokens (Encrypted)

Entity Name	Attributes	Key Type	Description
ERPTransaction	`transaction_id` (UUID)	**PK**	A specific record in ERP
	`erp_connection_id` (UUID)	**FK**	Links to **ERPConnection**
	`external_erp_id` (VARCHAR)		ID inside the ERP system
	`transaction_type` (VARCHAR)		e.g., "Invoice", "PO"
DocumentERPLink	`link_id` (UUID)	**PK**	Bridge table
	`document_id` (UUID)	**FK**	Links to **Document**
	`transaction_id` (UUID)	**FK**	Links to **ERPTransaction**
IntegrationMapping	`mapping_id` (UUID)	**PK**	Data transformation rules
	`erp_connection_id` (UUID)	**FK**	Links to **ERPConnection**
	`dms_field` (VARCHAR)		Source field in DMS
	`erp_field` (VARCHAR)		Destination field in ERP

Relationships:

- **ERPConnection (1) ↔ (N) ERPTransaction**: One connection handles many transactions.
- **Document (1) ↔ (N) DocumentERPLink ↔ (1) ERPTransaction**: A Many-to-Many relationship allowing one document to link to multiple ERP records (and vice versa).

4. Workflow & Automation

Entity Name	Attributes	Key Type	Description
WorkflowTemplate	`template_id` (UUID)	**PK**	Blueprint for a workflow
	`name` (VARCHAR)		e.g., "Invoice Approval"
	`trigger_event` (VARCHAR)		e.g., "Upload", "Update"
WorkflowInstance	`instance_id` (UUID)	**PK**	A running workflow

Entity Name	Attributes	Key Type	Description
	`template_id` (UUID)	**FK**	Links to **WorkflowTemplate**
	`document_id` (UUID)	**FK**	Links to **Document**
	`status` (ENUM)		In Progress, Completed
WorkflowTask	`task_id` (UUID)	**PK**	A specific step
	`instance_id` (UUID)	**FK**	Links to **WorkflowInstance**
	`assigned_user_id` (UUID)	**FK**	Links to **User**
	`action` (ENUM)		Approve, Reject, Review

Relationships:

- **WorkflowTemplate (1) ↔ (N) WorkflowInstance**: A template spawns many instances.
- **WorkflowInstance (1) ↔ (N) WorkflowTask**: An instance consists of many tasks.

5. Security, Compliance & Audit

Entity Name	Attributes	Key Type	Description
AuditLog	`log_id` (UUID)	**PK**	Immutable record of action
	`user_id` (UUID)	**FK**	Who did it
	`action` (VARCHAR)		What they did
	`entity_id` (UUID)		Target object ID
	`timestamp` (TIMESTAMP)		Exact time
DigitalSignature	`signature_id` (UUID)	**PK**	E-signature record
	`document_version_id` (UUID)	**FK**	Links to **DocumentVersion**
	`signer_user_id` (UUID)	**FK**	Links to **User**
	`hash` (VARCHAR)		Cryptographic hash

6. Advanced Processing (AI/OCR)

Entity Name	Attributes	Key Type	Description
ProcessingJob	`job_id` (UUID)	**PK**	Background task
	`document_version_id` (UUID)	**FK**	Links to **DocumentVersion**
	`job_type` (ENUM)		OCR, Virus Scan, Embedding
	`status` (ENUM)		Pending, Failed, Success
Tag	`tag_id` (UUID)	**PK**	Label for documents
	`name` (VARCHAR)		e.g., "Confidential"
DocumentTag	`document_id` (UUID)	**FK**	Junction Table
	`tag_id` (UUID)	**FK**	Junction Table

Relationships:

- **DocumentVersion (1) ↔ (N) ProcessingJob**: One file version can trigger multiple jobs (OCR, Scan, etc.).
- **Document (N) ↔ (N) Tag**: Documents can have many tags, tags can belong to many documents

7. ERP Integration

Entity Name	Attributes	Type	Description
MasterDataCache	`entity_id` (PK)	UUID	Local copy of Vendor/Customer data.
	`erp_connection_id` (FK)	UUID	Links to **ERPConnection**.
	`external_id`	VARCHAR	ID in the ERP (e.g., Vendor ID).
	`type`	ENUM	Customer, Vendor, Product.
	`data`	JSON	Cached address/contact info.
	`last_synced_at`	TIMESTAMP	When cache was updated.
WebhookConfig	`webhook_id` (PK)	UUID	Config for real-time ERP events.

Entity Name	Attributes	Type	Description
	`erp_connection_id` (FK)	UUID	Links to **ERPConnection**.
	`event_type`	VARCHAR	e.g., "PO_CREATED".
	`callback_url`	VARCHAR	URL the ERP sends data to.
	`secret_key`	VARCHAR	For validating the webhook.

8. Security & Compliance

Entity Name	Attributes	Type	Description
SearchLog	`search_id` (PK)	UUID	Tracks user search behavior.
	`user_id` (FK)	UUID	Who searched.
	`query`	VARCHAR	What they typed.
	`clicked_doc_id` (FK)	UUID	Which result they chose.
	`timestamp`	TIMESTAMP	Time of search.

Relationships

- **ERPConnection** `(1)` to `(0..N)` **MasterDataCache**
- **ERPConnection** `(1)` to `(0..N)` **WebhookConfig**
- **User** `(1)` to `(0..N)` **SearchLog**